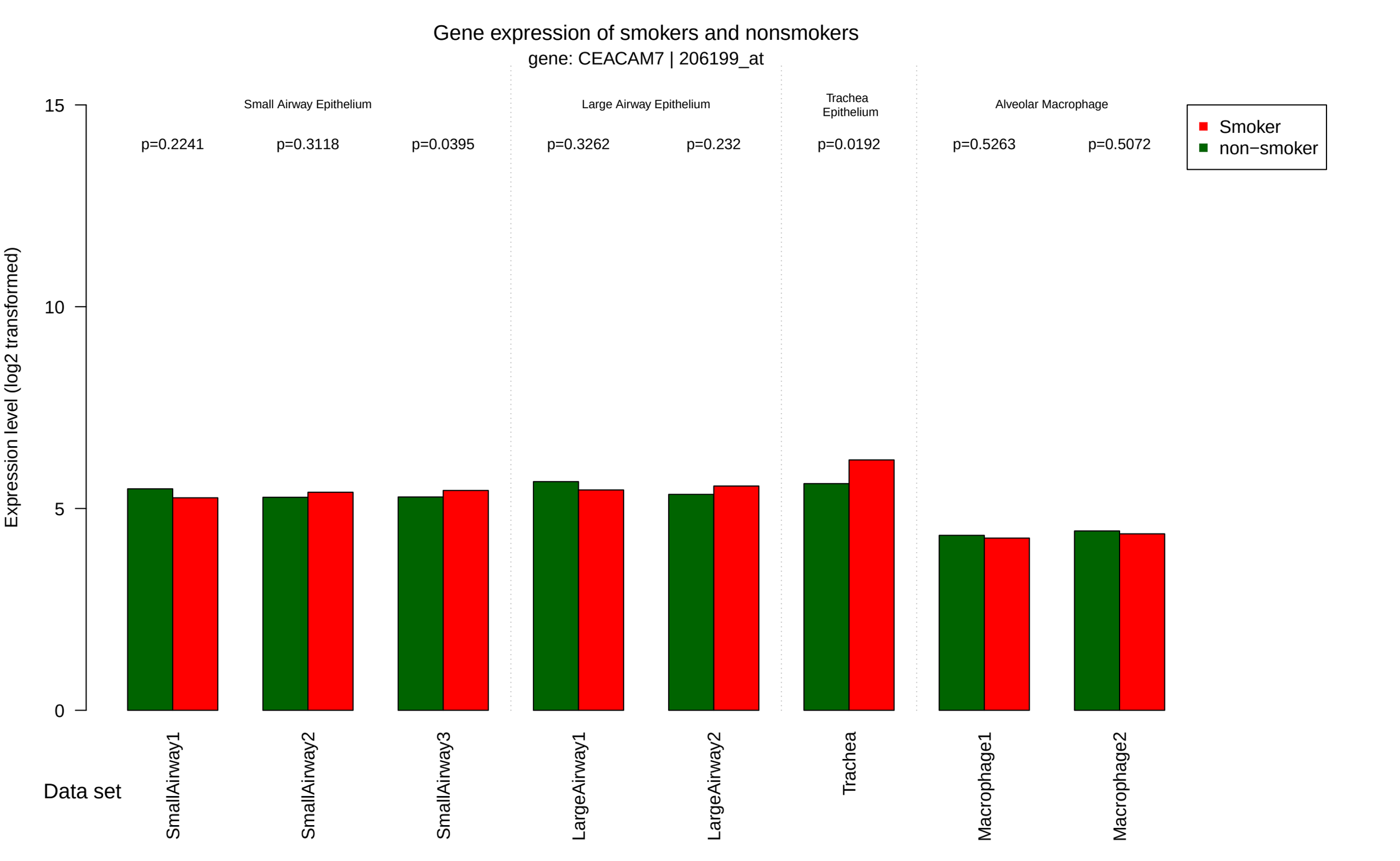
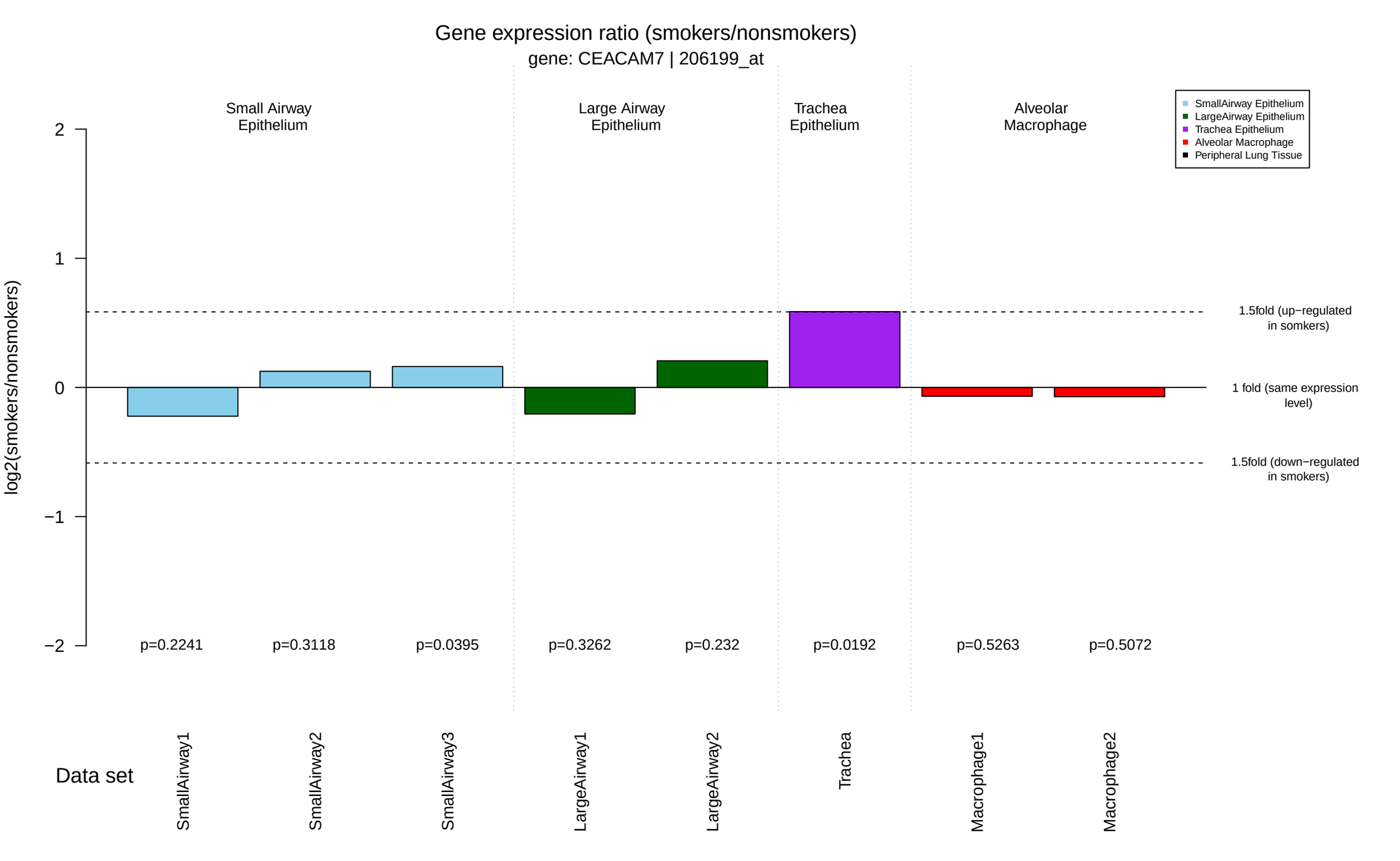
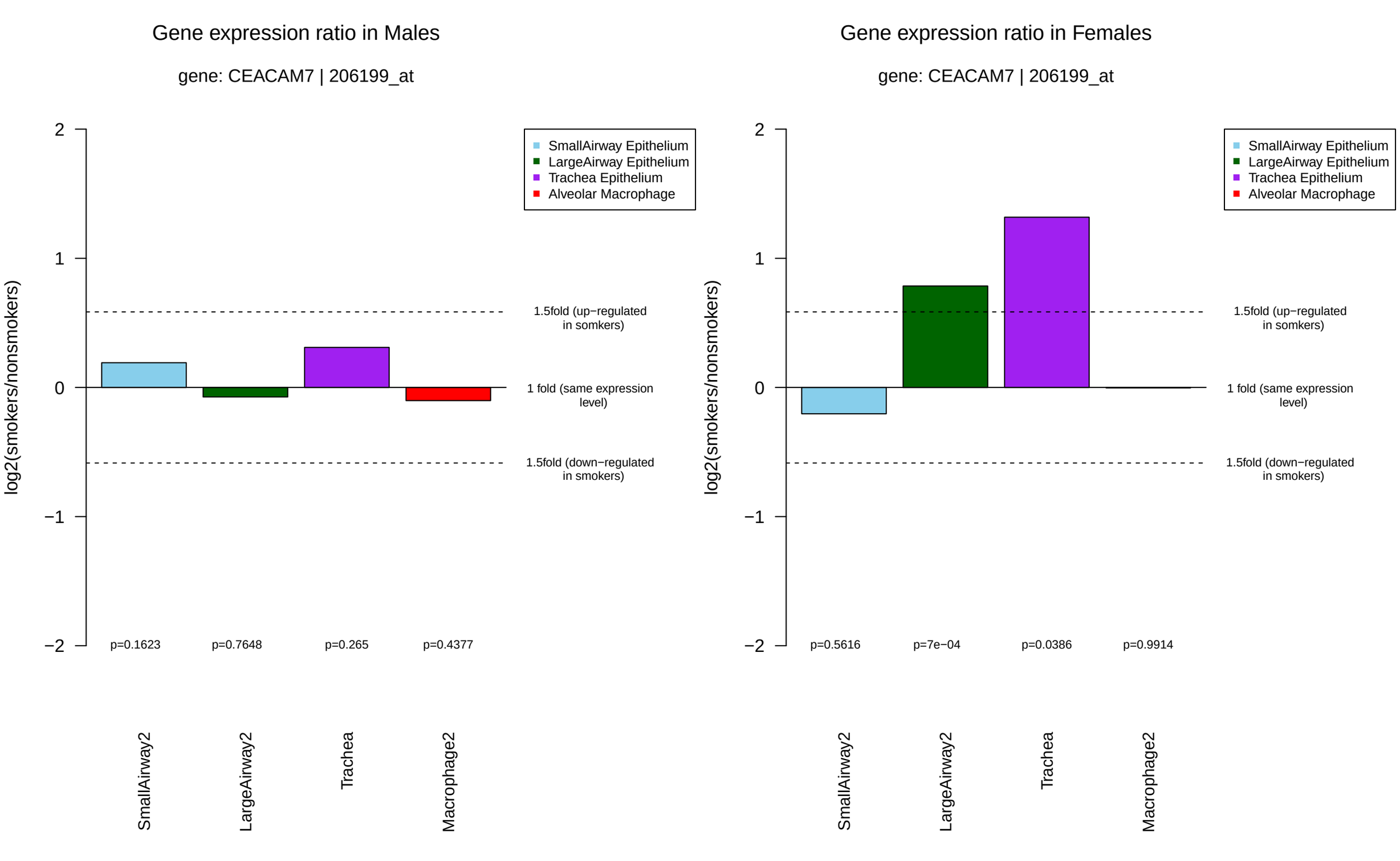
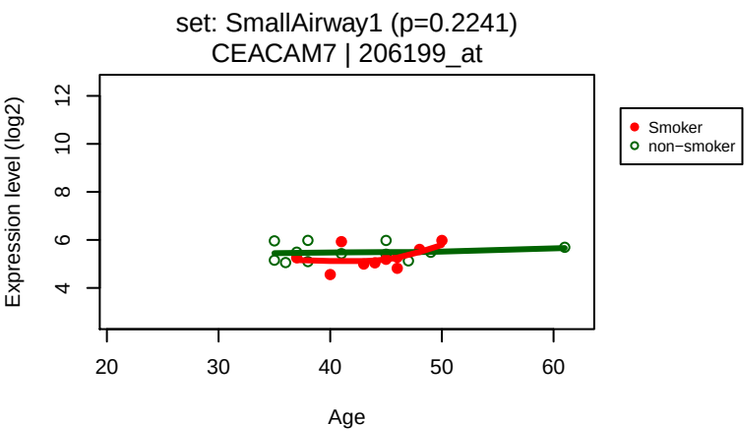
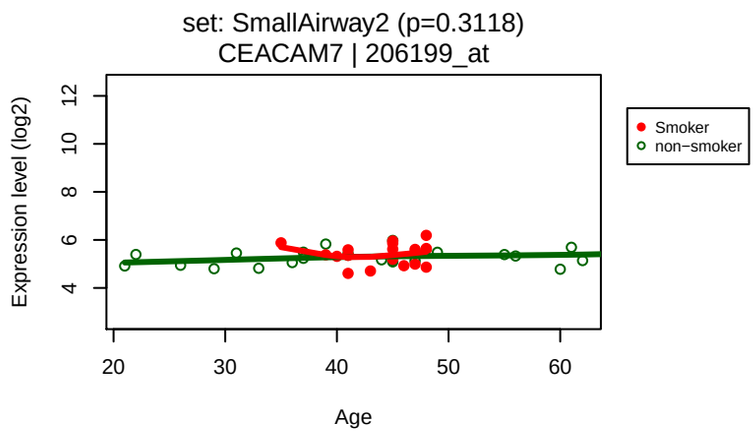
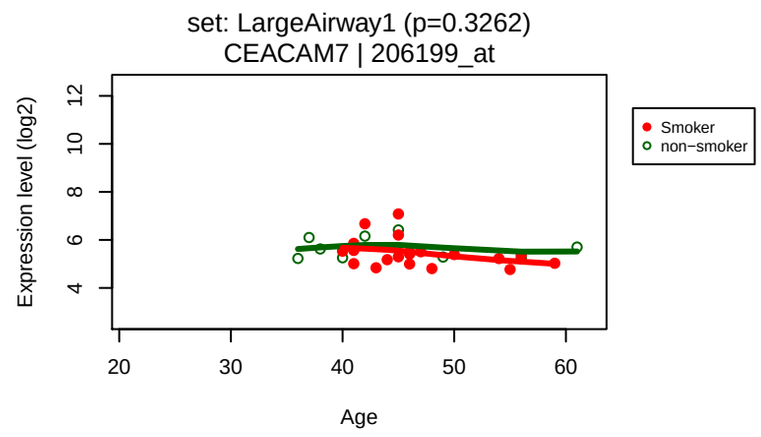

Gene expression of smokers and nonsmokers
gene: CEACAM7 | 206199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2241
p=0.3118
p=0.0395
p=0.3262
p=0.232
p=0.0192
p=0.5263
p=0.5072
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

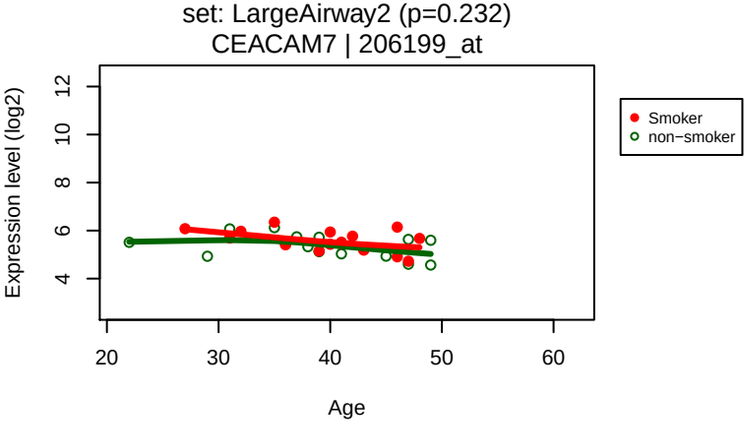

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM7 | 206199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2241
p=0.3118
p=0.0395
p=0.3262
p=0.232
p=0.0192
p=0.5263
p=0.5072
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

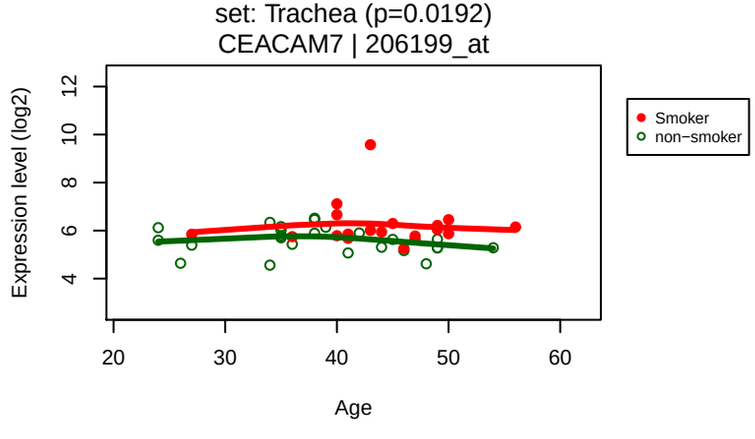

Gene expression ratio in Males
gene: CEACAM7 | 206199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1623
p=0.7648
p=0.265
p=0.4377
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM7 | 206199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5616
p=7e−04
p=0.0386
p=0.9914
SmallAirway2
LargeAirway2
Trachea
Macrophage2

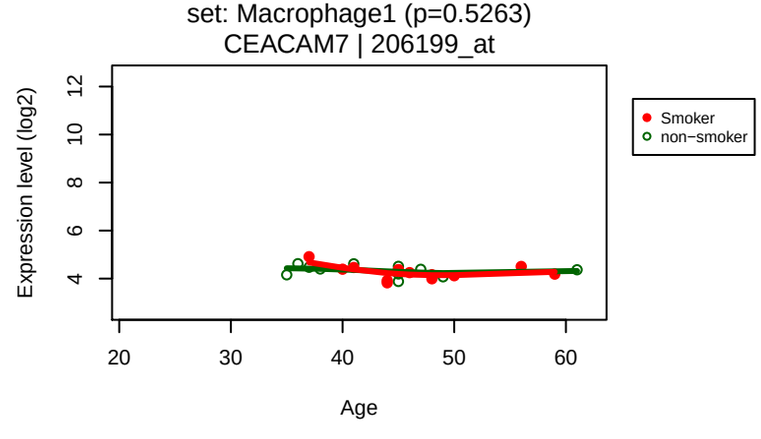

set: SmallAirway1 (p=0.2241)
CEACAM7 | 206199_at
Expression level (log2)
Age
Smoker
non−smoker

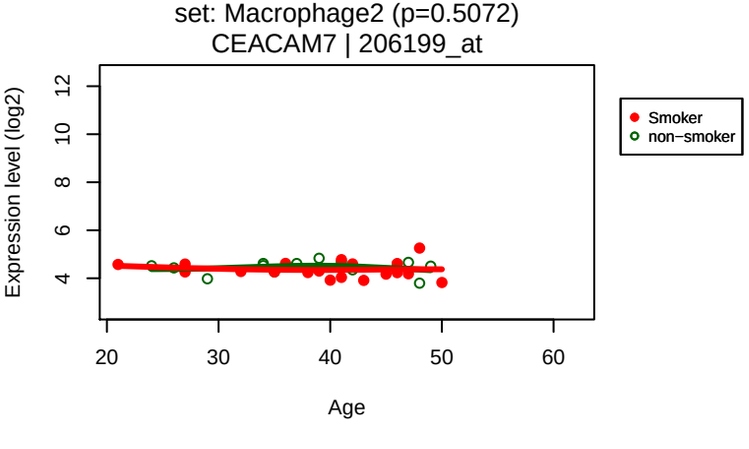

set: SmallAirway2 (p=0.3118)
CEACAM7 | 206199_at
Expression level (log2)
Age
Smoker
non−smoker

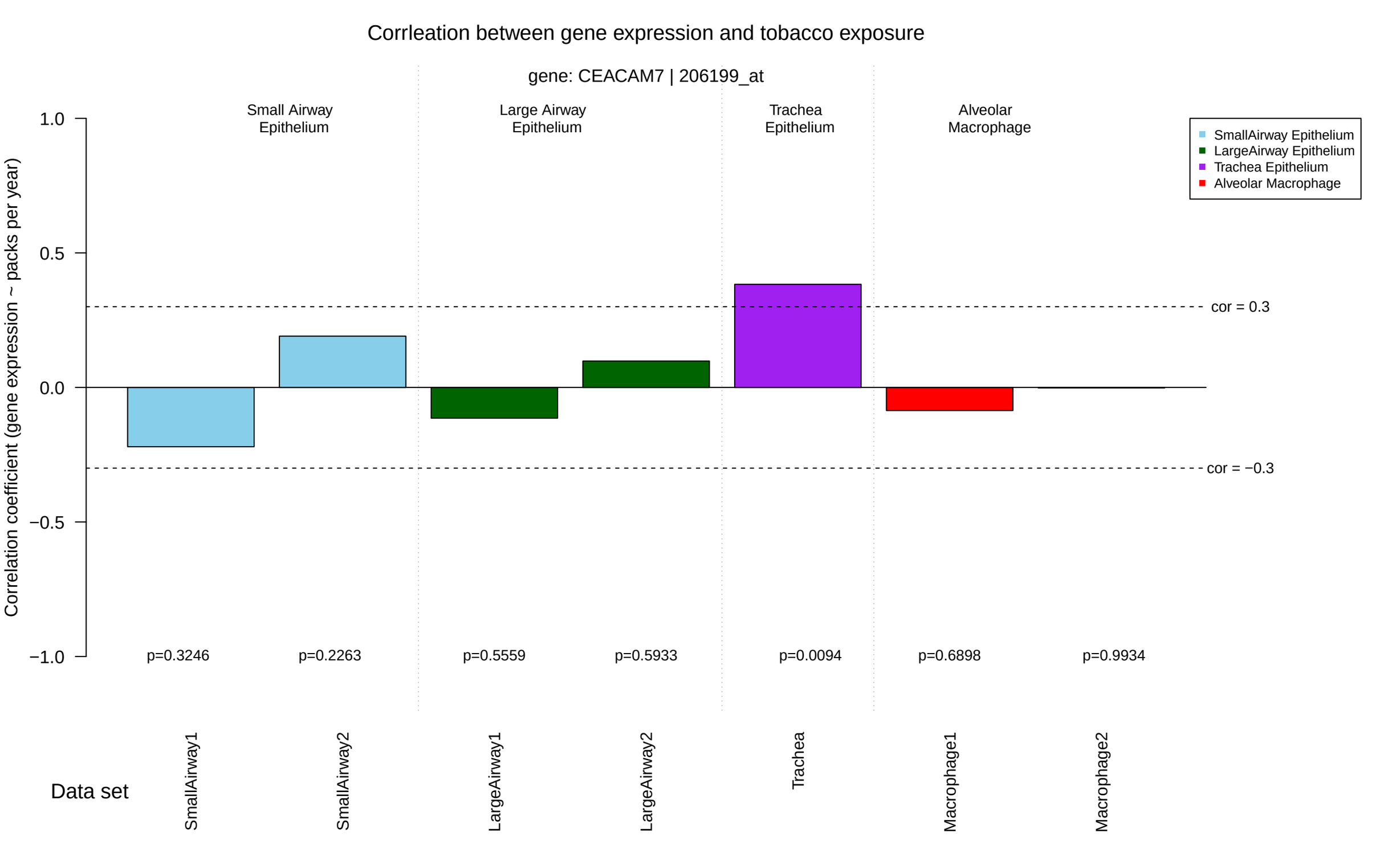

set: LargeAirway1 (p=0.3262)
CEACAM7 | 206199_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.232)
CEACAM7 | 206199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0192)
CEACAM7 | 206199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5263)
CEACAM7 | 206199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5072)
CEACAM7 | 206199_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM7 | 206199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3246
p=0.2263
p=0.5559
p=0.5933
p=0.0094
p=0.6898
p=0.9934
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2